Results log

Time series research harness — Part 1: evaluation harness and baselines

Generated October 12, 2026

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1 How this document works

This document records every result of the project: the question a test was meant to answer, how it was run, what came out, and what was concluded or decided from it. Design decisions

themselves live in `DECISIONS.md` and `decisions/` (cited as Dn); open questions for supervision in `QUESTIONS.md` (Qn).

Entries. Each result is an entry Rn, numbered in the order the runs were made. Numbers are never reused. An entry has: question, test, result, finding and consequence, plus open points where something is unresolved.

Status.

- **Current**: the latest result for its question.
- **Superseded by Rm** (kept): a later entry answers the same question, but this one is kept because it explains a decision or shows something the later one does not. Kept entries move to Appendix A.
- Superseded and replaced: a plain rerun with nothing to add. The text is removed; the entry number and its replacement stay in Table 1.

Which status an entry gets is decided by the user, entry by entry.

Numbers. All tables, figures and numbers quoted in the text are generated by `scripts/make_report.py` from the run folders (`runs/`, not committed); Appendix B lists the folder behind each entry. Unless stated otherwise: mean ± standard deviation over seeds (sample SD, $n-1$); the test set is scored once; accuracies are fractions, not percentages.

Reproduction criterion (D26 item 6, also used in D38). Implemented in `scripts/evaluate.py` and `scripts/tfc_official.py` as

$$|\text{published} - \text{mean}| \;\le\; \max(2\,\text{SD},\; 0.01).$$

Table 1: All entries.

Entry	Date	Question	Status
R1	2026-10-07	Do the data conversions reproduce the source values?	**Current**
R2	2026-10-07	Does TS2Vec reproduce on FordA? (CPU check)	Superseded by R3 (kept)
R3	2026-10-08	Does TS2Vec reproduce on FordA?	**Current**
R4	2026-10-08	Does T-Loss reproduce on FordA?	**Current**
R5	2026-10-08	Does the official TF-C code reproduce FD-A → FD-B (full data)?	**Current**
R6	2026-10-08	Does our TF-C reproduce FD-A → FD-B?	**Current**
R7	2026-10-08	Did our TF-C's pretraining learn a useful representation?	**Current**
R8	2026-10-08	Can FD-B separate methods at all?	**Current**
R9	2026-10-08	Does the official TF-C reproduce with its debugging subset?	**Current**
R10	2026-10-09	Where in frequency do the Herning AE and US signals carry information?	**Current**

Table 2: Reproduction checks: published accuracy against our mean over seeds. Criterion as implemented in `scripts/evaluate.py` and `scripts/tfc_official.py` (D26, D38): pass if $|\Delta| \le \max(2\,\text{SD}, 0.01)$.

Entry	Method	Data	Published	Ours (mean ± SD)	$\lvert\Delta\rvert$	Allowed	Criterion
R2	TS2Vec (CPU check)	FordA	0.9360	0.932 ± 0.003 ($n=3$)	0.004	0.010	pass
R3	TS2Vec	FordA	0.9360	0.932 ± 0.002 ($n=3$)	0.004	0.010	pass
R4	T-Loss	FordA	0.9280	0.933 ± 0.005 ($n=3$)	0.005	0.010	pass
R5	TF-C, official, full data	FD	0.8934	no test result	–	–	–
R6	TF-C, ours	FD	0.8934	0.706 ± 0.112 ($n=5$)	0.187	0.224	pass
R9	TF-C, official, subset	FD	0.8934	0.382 ± 0.164 ($n=5$)	0.511	0.327	fail

2 Data layer

2.1 R1 Conversions reproduce the source values

Date	2026-10-07 – 2026-10-08
Status	**Current**
Decisions	D3, D5, D20–D25, D27, D30, D39, D44
Runs	none (unit tests; see below)

Question. Does the conversion to the canonical format (D3, D5) return exactly the values in the source files?

Test.

- Herning (`tests/test_herning.py`): the first 20 sweeps of `scope_20260901_112732.h5` are converted into a temporary folder; every sample of both channels (AE, UL) is compared with the source (absolute tolerance 10^{-9}, far below one code step), and a slice must equal the corresponding part of the full signal. The test also checks four tables, one telemetry row per capture, two channels and time order.
- FordA: converted values compared with the archive's text files.
- FD-A: checked for non-finite values before the TF-C divergence was investigated (D44).

Result. Herning conversion exact on the 20 sweeps tested; FordA exact against the files; FD-A contains no non-finite values. These outcomes are recorded in `HANDOFF.md` and D44; no test log is stored.

Finding. The data layer can be trusted for the runs below. The full Herning run has not yet been converted; the band check (R10) reads the source file directly.

Open: Test outcomes are not stored anywhere but in notes. Should test runs write a log (e.g. into `runs/`) so that this entry can cite one?

3 Public benchmark: FordA

FordA (UCR archive, used as distributed: D27, D30) is the first slice: a wrapped official method, frozen representations, two probes (D26), three seeds, label budgets 1 %, 10 % and 100 % (D29, D40). Its purpose is to validate the harness against published numbers before anything new is measured.

3.1 R3 TS2Vec reproduces on FordA

Date	2026-10-08
Status	**Current**
Decisions	D26, D27, D28, D29
Runs	`20261008_084722_ts2vec_forda`

Question. Does TS2Vec, wrapped from its official code with the official settings, reproduce the published FordA accuracy (0.936, batch size 8)?

Test. TS2Vec trained on the FordA training inputs (no labels) with the official defaults (D28; 600 iterations); one 320-dimensional representation per series, max-pooled over time. Probes: TS2Vec's own RBF-SVM protocol and the harness logistic regression (D26), each at three label budgets. Three seeds; GPU; committed code (commit `adf16c6`).

Result. SVM probe with all labels: 0.932 ± 0.002 (criterion: pass, via the 1-pp clause; Table 2). All probes and budgets in Table 3 and Figure 1.

Finding. The wrapped TS2Vec and the evaluation pipeline reproduce the published number.

Consequence. The first slice is validated; a second method on the same data followed (R4, D32).

3.2 R4 T-Loss reproduces on FordA; FordA does not separate the two methods

Date	2026-10-08
Status	**Current**
Decisions	D32 (D26, D27, D29)
Runs	`20261008_092321_tloss_forda`

Question. Does T-Loss, wrapped from its official code, reproduce its published FordA accuracy (0.928, $K = 10$)? And does FordA tell TS2Vec and T-Loss apart?

Test. As R3, with T-Loss's official settings (D32: 2,000 steps, $K = 10$, representation size 320). Same probes, label budgets and seeds. The working tree had uncommitted changes (Appendix B).

Result. SVM probe with all labels: 0.933 ± 0.005 (criterion: pass). At every probe and label budget the two methods lie within about one standard deviation of each other (Table 3, Figure 1).

Finding. T-Loss reproduces. On FordA the two methods are indistinguishable with three seeds.

Consequence. TF-C was not started on FordA: no published FordA number to validate it against, and FordA does not separate the methods already wrapped (rationale of D35).

Table 3: FordA test set, frozen representations: mean ± SD over 3 seeds (R3, R4). The label subset is redrawn for each seed (D29).

Method	Probe	Labels	n	Accuracy	Macro-F1
TS2Vec	SVM (TS2Vec protocol)	1%	36	0.877 ± 0.022	0.877 ± 0.022
TS2Vec	SVM (TS2Vec protocol)	10%	360	0.914 ± 0.006	0.914 ± 0.006
TS2Vec	SVM (TS2Vec protocol)	100%	3601	0.932 ± 0.002	0.931 ± 0.002
TS2Vec	Logistic regression	1%	36	0.864 ± 0.016	0.864 ± 0.016
TS2Vec	Logistic regression	10%	360	0.895 ± 0.016	0.895 ± 0.016
TS2Vec	Logistic regression	100%	3601	0.923 ± 0.002	0.923 ± 0.002
T-Loss	SVM (TS2Vec protocol)	1%	36	0.872 ± 0.021	0.872 ± 0.020
T-Loss	SVM (TS2Vec protocol)	10%	360	0.916 ± 0.002	0.916 ± 0.002
T-Loss	SVM (TS2Vec protocol)	100%	3601	0.933 ± 0.005	0.933 ± 0.005
T-Loss	Logistic regression	1%	36	0.870 ± 0.022	0.870 ± 0.021
T-Loss	Logistic regression	10%	360	0.910 ± 0.008	0.909 ± 0.008
T-Loss	Logistic regression	100%	3601	0.926 ± 0.009	0.926 ± 0.009

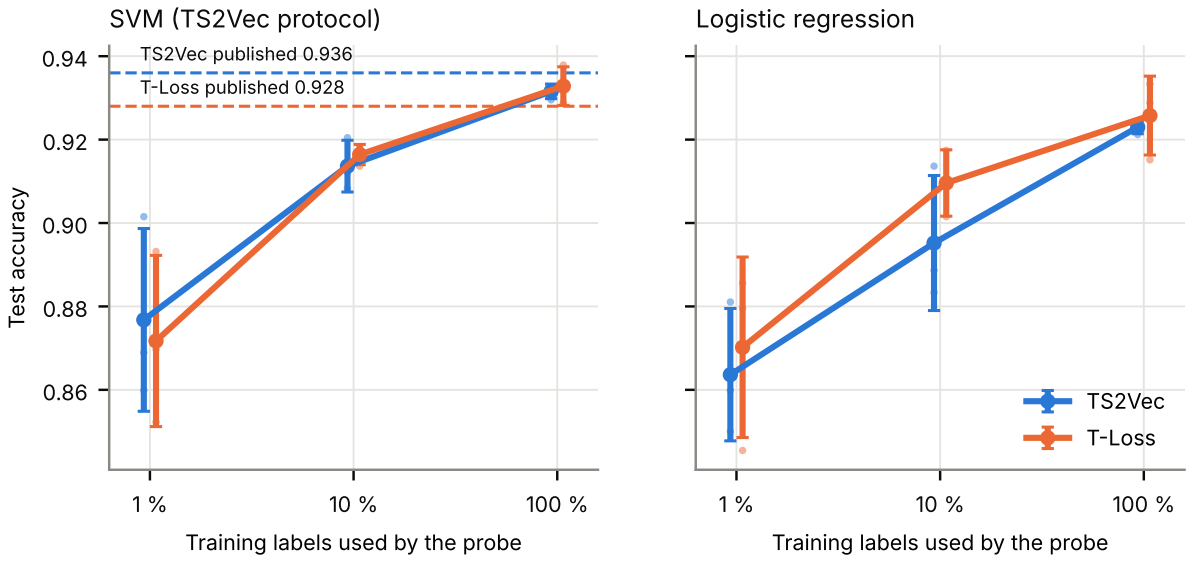


Figure 1: FordA test accuracy against the share of training labels given to the probe (R3, R4). Line and error bars: mean ± SD over 3 seeds; small dots: single seeds; dashed: published accuracies (SVM protocol only).

4 TF-C on FD-A → FD-B

TF-C is studied in steps (D35): (1) reproduce the published result with the official code; (2) the official TF-C on the full, raw Paderborn data; (3) our own corrected TF-C; (4) possibly a comparison on FordA. FD-A and FD-B are used as distributed by the TF-C authors (D39): pretraining on FD-A (train), fine-tuning on the 60 FD-B training windows, epoch selection on the 21 FD-B validation windows, testing on FD-B test. Target: 0.8934 ± 0.0379 accuracy, 5 runs (TF-C paper, Table 5; D38). The official reference is the 2023 code (D36), run with two protocols: *official* (epoch with the best *test* accuracy, as the official code) and *clean* (epoch with the best validation macro-F1, D37, D43).

4.1 R5 Official TF-C diverges on the full FD-A data

Date	2026-10-08
Status	**Current**
Decisions	D36, D37, D38, D44
Runs	`20261008_134951_tfc_official_fd`, `20261008_143021_tfc_official_fd`, `20261008_155516_tfc_official_fd`

Question. Does the official TF-C code, run on all FD-A windows (debugging subset off, D37), reproduce the published FD-A → FD-B accuracy?

Test. Official 2023 code through our adapter, settings checked against the official `FD_A_Configs.py` (40 epochs, batch 64, learning rate $3 \cdot 10^{-4}$, weight decay $3 \cdot 10^{-4}$ as hard-coded). Five seeds. Three attempts: the first (seed 0 only) and second with commit `87e9222`; the third with commit `c11957a`, which records a non-finite pretraining loss as divergence and skips fine-tuning for that seed (D44).

Result. In 4 of 5 seeds the pretraining loss became non-finite (Table 4, Figure 4); in the first attempt fine-tuning then crashed. The pretraining losses of the second and third attempt are identical for every seed, and the first attempt's finite losses (seed 0) are identical to them, so the divergence is deterministic given the seed. Seed 4 completed pretraining, but no attempt wrote fine-tuning output for it, and the third attempt wrote no `results.csv` (results file written: no).

Finding. On the full data, the official code does not produce a single test result. A likely cause, not verified (D44): gradients are reset only once per epoch, so each step on the full data uses many accumulated batches.

Consequence. D44: divergence is now recorded per seed as a result, and the official run was repeated with the debugging subset (R9).

Open: Why seed 4, whose pretraining finished, has no fine-tuning output, is not recorded in the run folder.

4.2 R6 Our TF-C falls well below the published accuracy

Date	2026-10-08
Status	**Current**
Decisions	D41, D42 (D38, D39)
Runs	`20261008_142751_tfc_fd`

Table 4: Official TF-C, full FD-A pretraining data (R5, third attempt). Pretraining loss per epoch is empty in `pretrain_loss.csv` once it is no longer finite.

Seed	Epochs run	Finite	First non-finite	Last finite loss	Fine-tuning output
0	5	4	5	8.290	no
1	22	21	22	8.186	no
2	17	16	17	8.190	no
3	5	4	5	8.340	no
4	40	40	–	8.185	no

Second and third attempt: pretraining losses identical for every seed.

Question. Does our TF-C (paper's CNN and loss, corrected frequency augmentation, selection by validation macro-F1; D41, D42) reach the published FD-A → FD-B accuracy?

Test. Settings in D41/D42 and the run's `config.yaml`. Five seeds. Component checks in `tests/test_tfc.py` (NT-Xent and L_C equal the official functions; frequency augmentation symmetric and the spectrum of a real signal; permutation reaches M segments); their outcome is not stored with the run. A second run folder with the same config (`20261008_143011_tfc_fd`) contains only the config and commit.

Result. Test accuracy 0.706 ± 0.112, macro-F1 0.757 ± 0.089; single seeds range from 0.519 to 0.803 (Table 5). The validation macro-F1 moves in large steps between epochs (Figure 2); one validation window is 1/21 of the validation set.

Finding. The mean is far below the published 0.8934, with a large spread over seeds. Formally, the criterion of D26 gives "pass", because twice the SD exceeds the gap (Table 2).

Consequence. D45 and D46: diagnostics to tell "pretraining learned little" from "fine-tuning or selection failed" (R7) and to check whether FD-B can separate methods (R8).

Open: The criterion passes a mean far below the target because the spread is large. Is R6 a reproduction? Should the criterion get a condition on the spread (D26 item 6 speaks of the 1-pp clause only "if the spread is very small", while the code applies it always)?

Table 5: Our TF-C (D41, D42), FD-A → FD-B test set (R6). Epoch: fine-tuning epoch with the best validation macro-F1; the test set is scored once, for that epoch. Precision, recall and F1 are macro averages.

Seed	Epoch	Acc.	Prec.	Recall	F1	AUROC	AUPRC
0	26	0.803	0.857	0.839	0.847	0.765	0.738
1	40	0.769	0.844	0.831	0.808	0.831	0.812
2	33	0.519	0.638	0.648	0.614	0.732	0.712
3	18	0.690	0.764	0.769	0.745	0.847	0.821
4	30	0.750	0.752	0.817	0.769	0.786	0.758
Mean ± SD		0.706 ± 0.112	0.771 ± 0.088	0.781 ± 0.079	0.757 ± 0.089	0.792 ± 0.047	0.768 ± 0.047

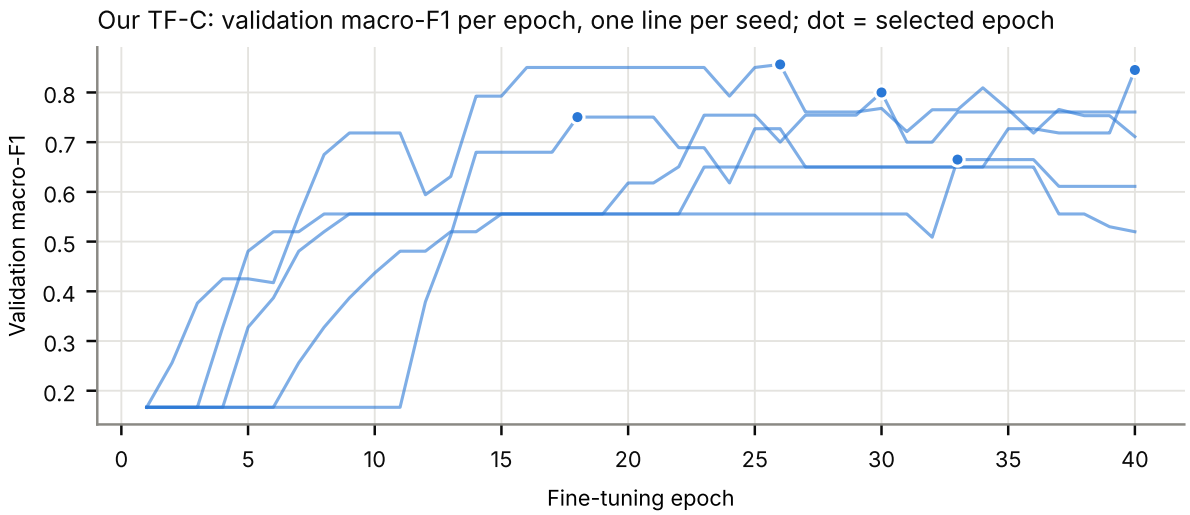


Figure 2: Our TF-C (R6): validation macro-F1 after every fine-tuning epoch, one line per seed; dot: the selected epoch (first maximum).

4.3 R7 Frozen probes on our TF-C encoders: pretraining is not where the shortfall is

Date	2026-10-08
Status	**Current**
Decisions	D45
Runs	`20261008_151014_probe_tfc_fd`

Question. Did our TF-C pretraining (R6) learn a representation that separates the FD-B classes, so that the low fine-tuned accuracy comes from fine-tuning or epoch selection? Or did pretraining learn little?

Test. The five pretrained encoders of R6 are frozen; their representation $[z_T; z_F]$ is probed with both harness probes trained on all 60 FD-B training labels and scored once on FD-B test (D45). Reference: untrained encoders of the same architecture, initialised with the same seeds, probed the same way. 100 % labels only (exception to D40: 1 % of 60 is less than one window).

Result. Pretrained encoder: SVM 0.951 ± 0.021, logistic regression 0.961 ± 0.012. Untrained encoder: SVM 0.973 ± 0.020, logistic regression 0.738 ± 0.091 (Table 7, Figure 3).

Finding. Probes on the frozen pretrained encoders score far above the fine-tuned TF-C (R6), which points to fine-tuning or epoch selection rather than pretraining. But pretraining does not beat an untrained encoder with the SVM probe; it helps only the linear probe.

4.4 R8 Plain spectra separate FD-B almost perfectly

Date	2026-10-08
Status	**Current**
Decisions	D46
Runs	`20261008_162447_probe_spectrum_fd`

Question. Can FD-B separate methods at all, given that an untrained encoder with an SVM probe already reaches high accuracy (R7)?

Test. Both harness probes trained directly on the full FFT magnitude spectrum of each FD-B training window (the input of TF-C's frequency branch), no encoder; 60 labels; scored once on FD-B test. Deterministic, so one value per probe. Run in Claude's cloud workspace on the work computer's FD-B data, copied unchanged (`NOTE.txt` in the run folder).

Result. SVM 0.999, logistic regression 0.943 (Table 7).

Finding. With 60 labels, an SVM on the plain spectrum classifies FD-B test almost without error, above the published TF-C result. Under this protocol FD-B cannot show that a pretrained method adds anything (the condition stated in D46's rationale is met).

Open: What follows for the TF-C study (D35) from R7–R9 has not been decided.

4.5 R9 Official TF-C with its debugging subset collapses to one class

Date	2026-10-08
Status	**Current**
Decisions	D37, D44 (D43)
Runs	`20261008_171159_tfc_official_fd_subset`

Question. Is the debugging subset what makes the official code trainable, and does it then reproduce the published accuracy (D37: first check if the reproduction fails; D44)?

Test. As R5, with the official debugging subset switched on (as shipped: 640 FD-A windows for pretraining). Five seeds; both protocols.

Result. No seed diverged (Figure 4). Test accuracy 0.382 ± 0.164, macro-F1 0.178 ± 0.069 (official protocol; criterion: fail). In every seed and every fine-tuning epoch, macro recall on test and validation is exactly 1/3 (Table 6), which is what predicting one class for every window gives. All epochs therefore tie and both protocols select epoch 1; official and clean numbers are identical.

Finding. With the subset the official code trains without diverging, but the fine-tuned model predicts a single class. Neither configuration of the official code reproduces the paper.

Table 6: Official TF-C with the debugging subset (R9). Official protocol: epoch with the best test accuracy; clean protocol: epoch with the best validation macro-F1 (D43). Last column: macro recall on test and validation is exactly 1/3 at every one of the 40 fine-tuning epochs.

Seed	Official protocol			Clean protocol			Recall = 1/3
	Epoch	Acc.	F1	Epoch	Acc.	F1	all epochs
0	1	0.455	0.209	1	0.455	0.209	yes
1	1	0.089	0.055	1	0.089	0.055	yes
2	1	0.455	0.209	1	0.455	0.209	yes
3	1	0.455	0.209	1	0.455	0.209	yes
4	1	0.455	0.209	1	0.455	0.209	yes

FD-B results side by side

Table 7: FD-B test set: TF-C results and the diagnostics (R6–R9). Probes use all 60 FD-B training labels (D45, D46). The |FFT| probes are deterministic (one value).

Model / probe	Entry	Accuracy	Macro-F1	Seeds
TF-C paper (published)	–	0.893 ± 0.038	0.916 ± 0.083	5
Official TF-C, subset, fine-tuned	R9	0.382 ± 0.164	0.178 ± 0.069	5
Our TF-C, fine-tuned	R6	0.706 ± 0.112	0.757 ± 0.089	5
SVM on pretrained TF-C encoder	R7	0.951 ± 0.021	0.964 ± 0.015	5
log. reg. on pretrained TF-C encoder	R7	0.961 ± 0.012	0.971 ± 0.009	5
SVM on untrained encoder	R7	0.973 ± 0.020	0.980 ± 0.015	5
log. reg. on untrained encoder	R7	0.738 ± 0.091	0.793 ± 0.085	5
SVM on |FFT| spectrum	R8	0.999	0.999	1
log. reg. on |FFT| spectrum	R8	0.943	0.958	1

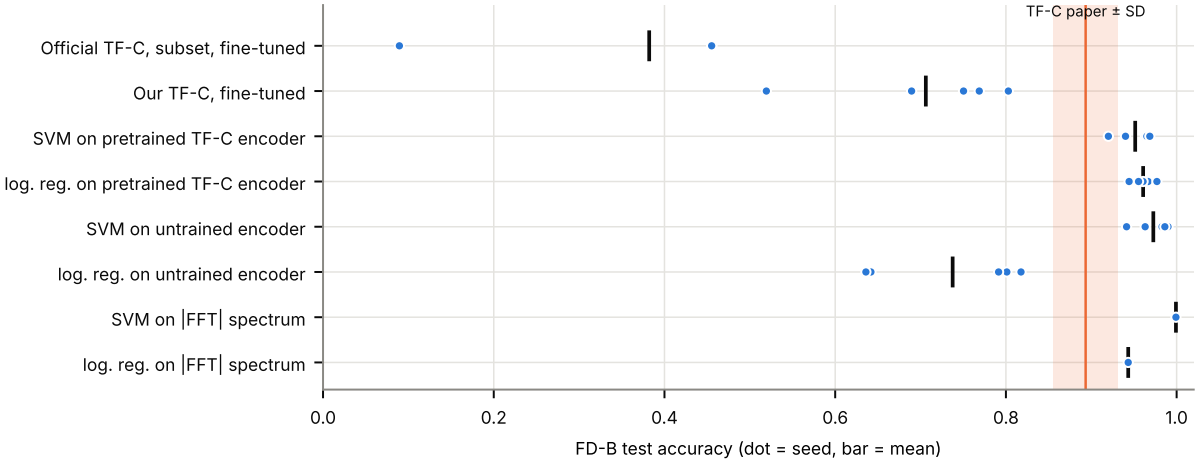


Figure 3: FD-B test accuracy for every TF-C model and probe (R6–R9): dots are seeds, bars means. Shaded: published TF-C accuracy ± SD.

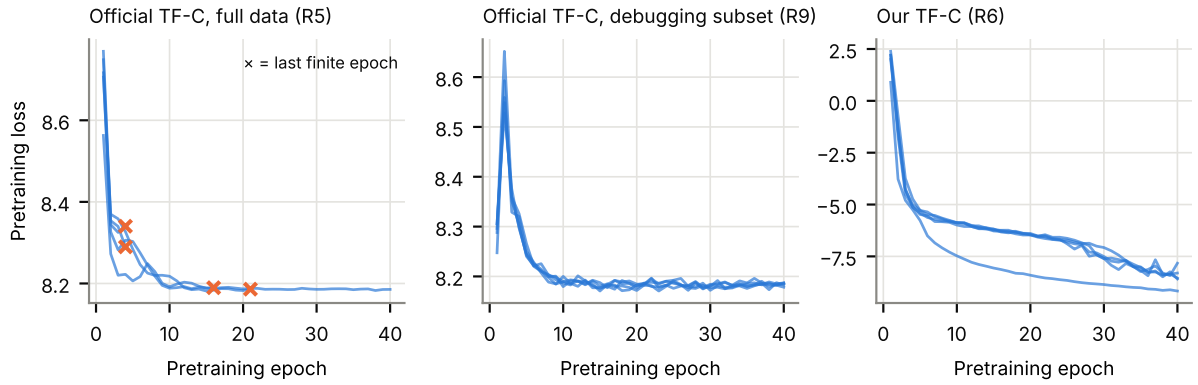


Figure 4: TF-C pretraining loss per epoch, one line per seed. Left: official code, full data (R5, third attempt); × marks the last finite epoch. Middle: official code, debugging subset (R9). Right: our TF-C (R6); its loss is defined differently (D41 item 4), so the scales are not comparable.

5 Herning test rig

The Herning run `scope_20260901_112732` is a validation run of an unfinished rig (D7, D22): one bearing, one lubricant, rpm swept up and down at 13 temperature targets. The regression target is κ, computed from rpm and temperature as in the MSSP code (D55). Figure 5 shows the operating points.

5.1 R10 Herning band check: where AE and US carry information

Date	2026-10-09
Status	**Current**
Decisions	D51, D52, D55; led to D60, D61, D62, Q3
Runs	`20261009_085307_band_check_herning`

Question. Before band-limiting and downsampling the rig signals (D51): in which frequency bands do AE and US carry signal from the rotating bearing, and does that signal depend on the operating point (D52)?

Test. `scripts/check_bands.py` on the source file. Per sweep and sensor: Welch power spectrum (Hann window, 65,536 samples, 50 % overlap; 38-Hz resolution at 2.5 MHz), integrated over a low band (0–891 Hz) and base-10 third-octave bands from 1 kHz to Nyquist. Sweeps: 3,964 in total; 3,704 analysed (rotating, D55 exclusions applied), 126 noise reference (rpm target 100: bearing not rotating), 129 at rpm target 0, 5 excluded (cool-down). Per band: ratio of the median power of analysed sweeps to the noise reference (dB), and Spearman correlation of band power with rpm, temperature and κ (κ from 0.045 to 1.651). Extra control against slow drift over the 13-hour run: the correlation with rpm computed within each temperature block (median over blocks). Run in four chunks of sweeps (3-minute limit per call), merged into one result (`NOTE.txt`). An earlier folder (`20261009_085150_band_check_herning`) contains only the config.

Result. Figure 6; all bands in Table 9 (Appendix C).

- AE: power above the reference from about 12.6 kHz to about 400 kHz, with peaks of 24.4 dB at 25.12 kHz and 24.0 dB at 100 kHz. In the bands from 12.6 to 398 kHz the within-block

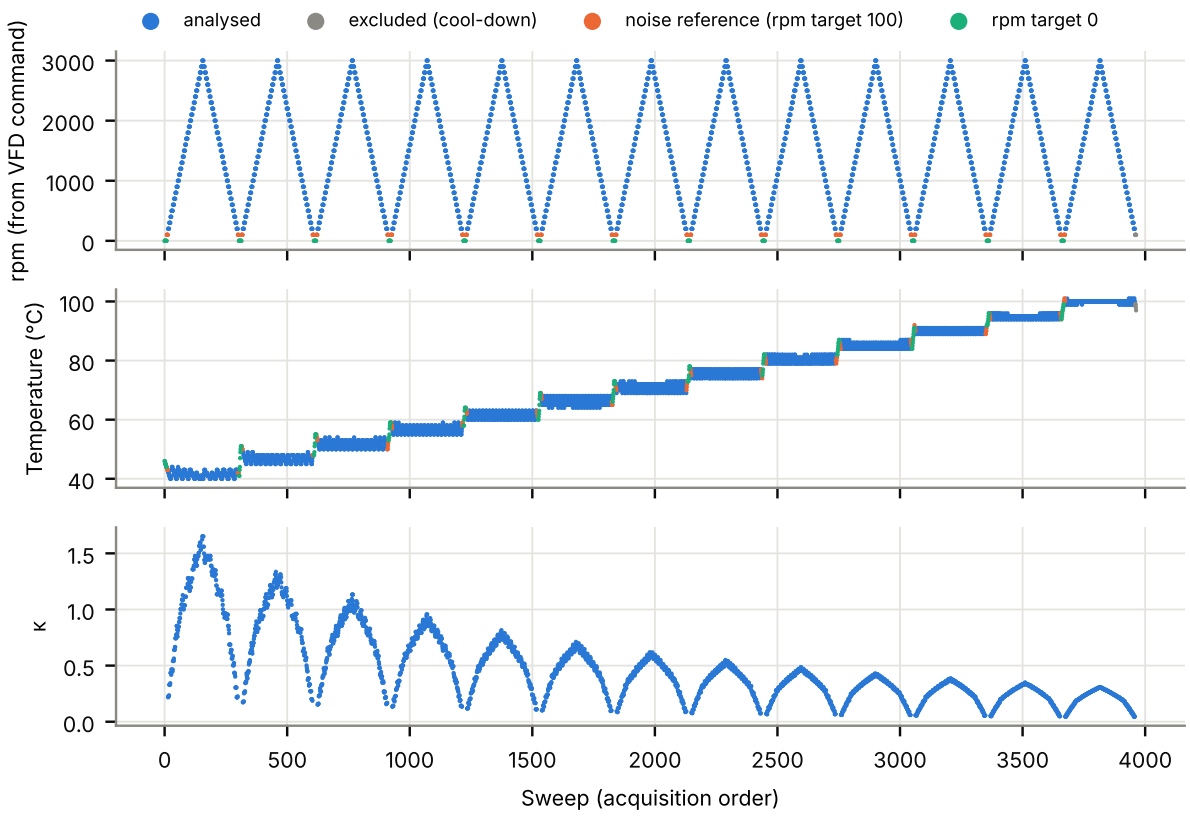


Figure 5: Herning run: rpm (from the frequency-converter command), temperature and κ for every sweep, in acquisition order, coloured by role in R10 (D52, D55).

correlation with rpm is 0.96–0.99. From 631 kHz up there is no excess power (at most −0.02 dB) and band power *falls* with rpm within blocks (ρ from −0.72 to −0.64).
- US: 35–49 dB above the reference up to 6.3 kHz. Between about 40 and 200 kHz the excess is only 0.6–2.6 dB, but band power still rises with rpm within blocks (ρ 0.62–0.85) and correlates with κ (ρ 0.53–0.81).

Finding. AE shows rpm-dependent power above the reference roughly between 12 and 560 kHz (the origin of the part below 50 kHz is unknown, D60); above about 560 kHz it shows no excess power, and the user regards it as noise (D61, D62). US carries most of its power below about 8 kHz, but rpm-dependent content continues up to about 200 kHz. Correlation within temperature blocks rules out slow drift, but not electrical interference that also scales with speed (Q3).

Consequence. Feature bands: AE 50–560 kHz (nothing below the 50-kHz coupler high-pass; D60, D61), US 0.71–178 kHz (D60). Downsampling: AE by 2 (1.25 MHz), US by 5 (500 kHz) (D62). Whether US above about 40 kHz is bearing signal or interference is open (Q3).

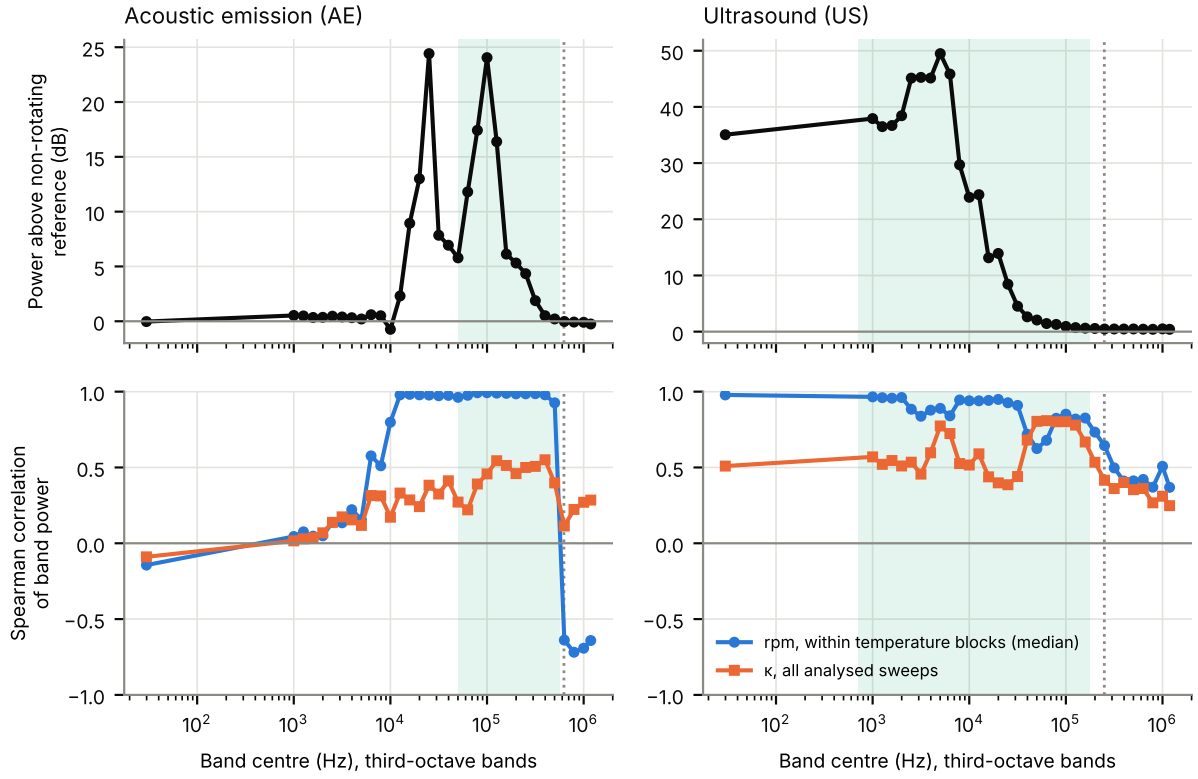


Figure 6: Herning band check (R10). Top: median band power of rotating sweeps relative to the non-rotating reference. Bottom: Spearman correlation of band power with rpm (median within temperature blocks) and with κ (all analysed sweeps). Shaded: feature bands (D60, D61). Dotted: Nyquist frequency after downsampling (D62). The leftmost point is the 0–891-Hz band.

6 Open points

From this document:

- R1: test outcomes are not stored as files.
- R5: why seed 4 of the official full-data run has no fine-tuning output.

- R6: whether R6 counts as a reproduction, and whether the reproduction criterion needs a condition on the spread.
- R8: what follows for the TF-C study (D35) from R7–R9.

For supervision (`QUESTIONS.md`): Q1 Herning split (temperature hold-out details); Q2 comparing whole-sweep features with window-based SSL representations; Q3 US above 40 kHz and AE above 560 kHz — bearing signal or electrical interference.

A Superseded entries kept for history

A.1 R2 TS2Vec on FordA, preliminary CPU check

Date 2026-10-07
Status **Superseded by R3** (kept)
Decisions D26, D27, D28, D29
Runs `20261007_214220_ts2vec_forda_cpu_check`

Question. Does the first slice run end to end and land near the published TS2Vec accuracy, before the GPU run on the work computer?

Test. As R3, run on CPU in Claude's cloud workspace from the uncommitted working tree (no commit hash).

Result. SVM probe with all labels within the criterion (pass; Table 2).

Finding. Kept because it shows the CPU run in the cloud workspace and the GPU run on the work computer agree, and because R3 was planned on its basis. Replaced by R3 as the result (committed code, GPU).

B Run provenance

Table 8: Run folders behind each entry. 'Uncommitted': the working tree had changes when the run started, so the commit alone does not reproduce the code.

Entry	Run folder (`runs/`)	Commit	Uncommitted	Device
R2	`20261007_214220_ts2vec_forda_cpu_check`	none	yes	cpu
R3	`20261008_084722_ts2vec_forda`	adf16c6	no	cuda
R4	`20261008_092321_tloss_forda`	5320251	yes	cuda
R5	`20261008_134951_tfc_official_fd`	87e9222	yes	cuda
R5	`20261008_143021_tfc_official_fd`	87e9222	yes	cuda
R5	`20261008_155516_tfc_official_fd`	c11957a	yes	cuda
R6	`20261008_142751_tfc_fd`	87e9222	yes	cuda
R7	`20261008_151014_probe_tfc_fd`	87e9222	yes	cuda
R8	`20261008_162447_probe_spectrum_fd`	none	–	–
R9	`20261008_171159_tfc_official_fd_subset`	c11957a	yes	cuda
R10	`20261009_085307_band_check_herning`	c11957a	yes	–
R6 (empty)	`20261008_143011_tfc_fd`	87e9222	yes	cuda
R10 (empty)	`20261009_085150_band_check_herning`	c11957a	yes	–

C Herning band check: all bands

Table 9: Herning band check, all bands (R10). Ratio: median band power of analysed sweeps over that of the non-rotating reference. ρ: Spearman correlation of band power with rpm (all analysed sweeps, and median within temperature blocks), temperature and κ.

Sensor	Band	Ratio (dB)	ρ rpm	ρ rpm in block	ρ temp.	ρ κ
AE	0-891.25 Hz	−0.0	−0.08	−0.14	0.13	−0.09
AE	1 kHz	0.5	0.03	0.04	0.10	0.02
AE	1.259 kHz	0.5	0.06	0.08	0.12	0.03
AE	1.585 kHz	0.3	0.04	0.05	0.08	0.04
AE	1.995 kHz	0.4	0.07	0.05	0.07	0.07
AE	2.512 kHz	0.5	0.16	0.14	0.04	0.14
AE	3.162 kHz	0.4	0.16	0.13	−0.01	0.17
AE	3.981 kHz	0.3	0.20	0.22	0.04	0.15
AE	5.012 kHz	0.2	0.08	0.16	0.00	0.12
AE	6.31 kHz	0.6	0.55	0.58	0.09	0.31
AE	7.943 kHz	0.5	0.48	0.51	0.05	0.31
AE	10 kHz	−0.7	0.70	0.80	0.30	0.17
AE	12.59 kHz	2.3	0.90	0.98	0.29	0.33
AE	15.85 kHz	8.9	0.88	0.98	0.35	0.29
AE	19.95 kHz	13.0	0.86	0.98	0.40	0.24
AE	25.12 kHz	24.4	0.90	0.98	0.24	0.38
AE	31.62 kHz	7.8	0.87	0.97	0.29	0.32
AE	39.81 kHz	6.9	0.91	0.98	0.20	0.41
AE	50.12 kHz	5.8	0.86	0.96	0.35	0.27
AE	63.1 kHz	11.8	0.85	0.98	0.42	0.22
AE	79.43 kHz	17.4	0.95	0.99	0.25	0.39
AE	100 kHz	24.0	0.96	0.99	0.17	0.46
AE	125.9 kHz	16.4	0.98	0.99	0.07	0.54
AE	158.5 kHz	6.1	0.97	0.99	0.11	0.51
AE	199.5 kHz	5.3	0.96	0.99	0.17	0.46
AE	251.2 kHz	4.3	0.97	0.99	0.12	0.50
AE	316.2 kHz	1.9	0.96	0.99	0.11	0.51
AE	398.1 kHz	0.5	0.97	0.98	0.05	0.55
AE	501.2 kHz	0.2	0.83	0.93	0.16	0.40
AE	631 kHz	−0.0	−0.43	−0.64	−0.34	0.11
AE	794.3 kHz	−0.1	−0.33	−0.72	−0.46	0.22
AE	1000 kHz	−0.1	−0.27	−0.69	−0.49	0.27
AE	1259 kHz	−0.2	−0.25	−0.64	−0.50	0.28
US	0-891.25 Hz	35.0	0.96	0.98	0.09	0.51
US	1 kHz	37.9	0.96	0.97	0.02	0.57
US	1.259 kHz	36.5	0.93	0.96	0.07	0.52
US	1.585 kHz	36.7	0.94	0.96	0.04	0.55
US	1.995 kHz	38.4	0.95	0.96	0.09	0.51
US	2.512 kHz	45.1	0.85	0.88	−0.00	0.53
US	3.162 kHz	45.3	0.82	0.84	0.07	0.46
US	3.981 kHz	45.1	0.84	0.88	−0.09	0.60
US	5.012 kHz	49.5	0.81	0.89	−0.31	0.77
US	6.31 kHz	45.8	0.78	0.84	−0.26	0.72
US	7.943 kHz	29.7	0.91	0.94	0.06	0.53
US	10 kHz	23.9	0.91	0.94	0.06	0.52
US	12.59 kHz	24.4	0.90	0.94	−0.03	0.59
US	15.85 kHz	13.1	0.88	0.94	0.15	0.44
US	19.95 kHz	13.9	0.88	0.95	0.21	0.40
US	25.12 kHz	8.4	0.85	0.93	0.21	0.39

Sensor	Band	Ratio (dB)	ρ rpm	ρ rpm in block	ρ temp.	ρ κ
US	31.62 kHz	4.5	0.85	0.91	0.14	0.44
US	39.81 kHz	2.6	0.64	0.72	−0.25	0.68
US	50.12 kHz	2.1	0.60	0.62	−0.42	0.80
US	63.1 kHz	1.4	0.63	0.68	−0.42	0.81
US	79.43 kHz	1.3	0.76	0.82	−0.35	0.80
US	100 kHz	0.9	0.78	0.85	−0.35	0.80
US	125.9 kHz	0.7	0.74	0.82	−0.36	0.78
US	158.5 kHz	0.6	0.79	0.83	−0.21	0.67
US	199.5 kHz	0.6	0.71	0.73	−0.11	0.53
US	251.2 kHz	0.5	0.64	0.64	−0.01	0.42
US	316.2 kHz	0.5	0.53	0.50	−0.03	0.36
US	398.1 kHz	0.5	0.40	0.41	−0.15	0.40
US	501.2 kHz	0.5	0.37	0.41	−0.11	0.35
US	631 kHz	0.4	0.40	0.42	−0.10	0.36
US	794.3 kHz	0.4	0.37	0.37	−0.01	0.27
US	1000 kHz	0.5	0.51	0.51	0.04	0.31
US	1259 kHz	0.4	0.38	0.37	0.01	0.25